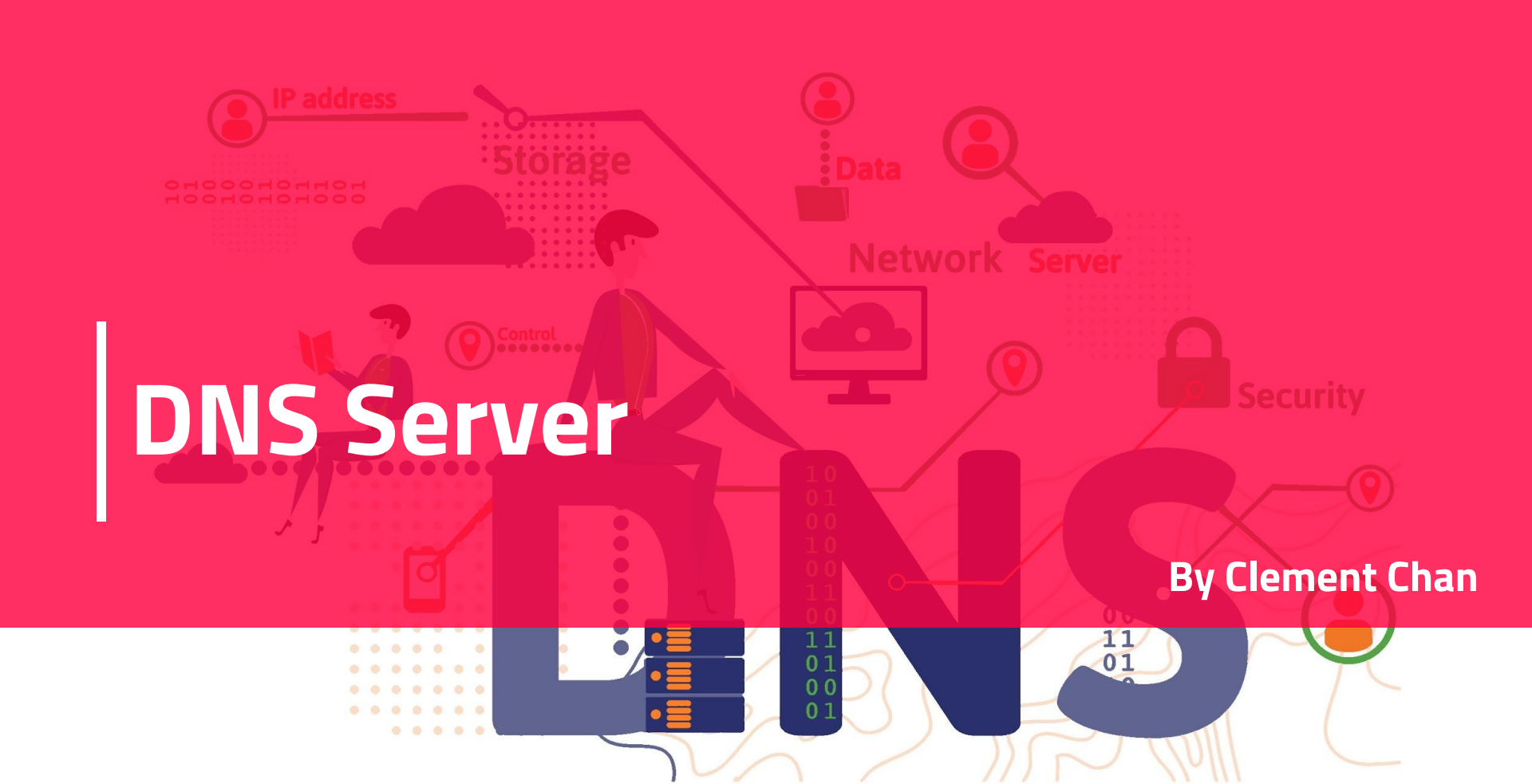
DNS Server



By Clement Chan

> *The Domain Name Server (DNS) is the Achilles heel of the Web. The important thing is that it's managed responsibly.*

\- Tim Berners-Lee

What is DNS?

Christo’s nickname

D - If you mess it up,
N - you
S - screwed

DNS
Domain Name System

- Translates domain names and IP addresses
- Backbone of pretty much everything
- Most servers depend on DNS to function properly

DNS Common Implementations

- BIND

Berkeley's, de-facto for Unix, more common

- MS DNS

Microsoft's, de-facto for Windows, core component of AD

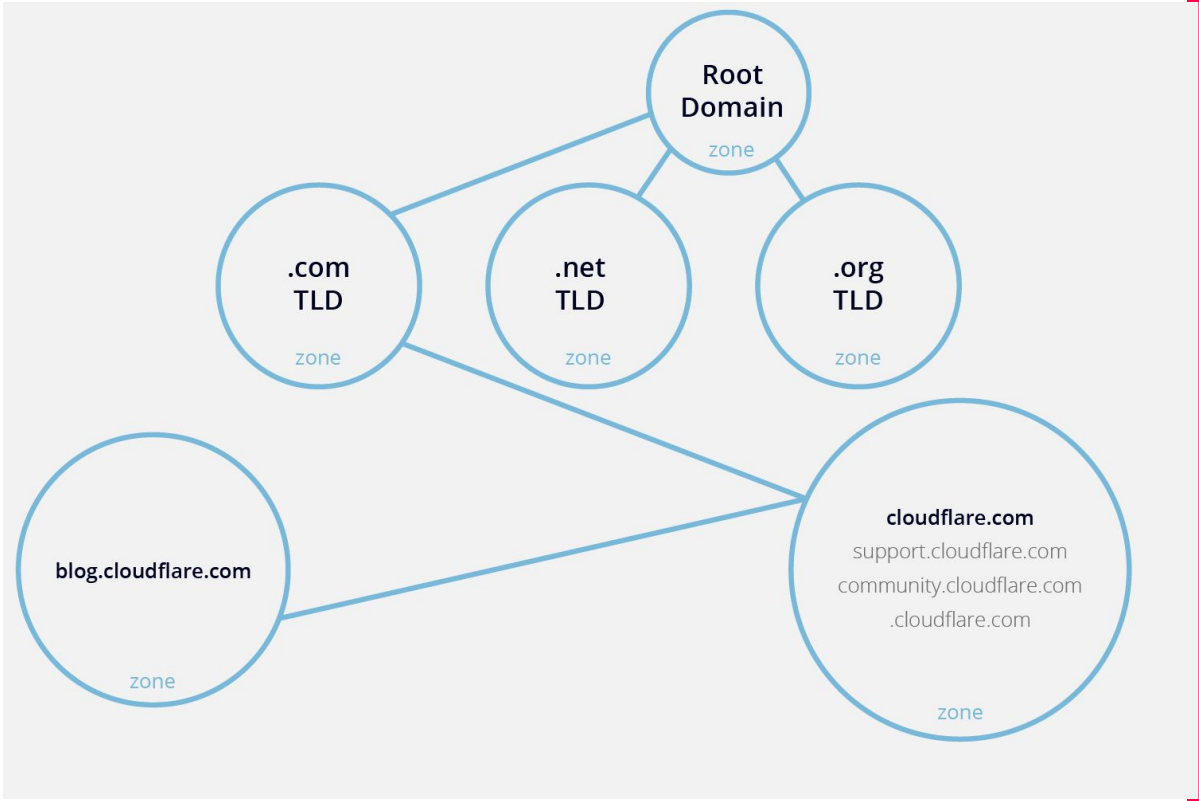
Some Terminology

- Zone

Portion of domain name space

- Zone transfer

Replicates DNS databases across DNS servers



DNS Dynamic Updates

Updates DNS records automatically when a new device joins the network or changes its IP address

Lessens administrative load

Home networks are under DHCP

Dynamic DNS allows you to link your changing home IP address to a custom domain name

You can use your hostname instead of your IP to connect to your home

DNS Name Server Types

All name servers store some type of DNS record

- Root

Manage requests for top-level domains

- Top-level

Contacts authoritative name servers

- Authoritative

Stores DNS records for domain names

- Recursive

Receives DNS queries and is responsible for finding an authoritative DNS server that can supply the IP address

Your computer and the recursive DNS server have local caches to speed up the process for common sites like Youtube or Google

DNS Query Types

- Recursive

Server will find IP on behalf of client, asking other DNS servers if it is not found in its local cache

- Iterative

Server will check local cache; if not found, it will refer the client to another DNS server

DNS Query Types (cont.)

- Forward DNS lookup

Find IP of a domain name

- Reverse DNS lookup

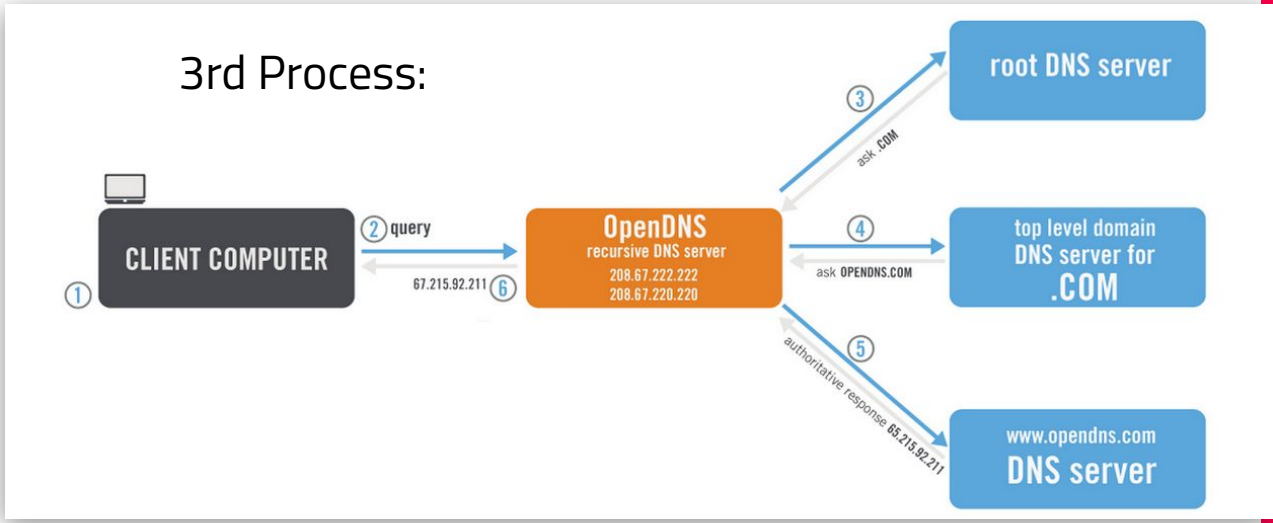
Find domain name of an IP

DNS Record Types

- **A** **Domain to IP**
- **CNAME** **Domain name alias/forwarding**
- **MX** **Mail routing**
- TXT Information purposes
- NS Authoritative nameserver identification
- SOA Administrative info for zone transfers
- SRV Service records
- **PTR** **IP to Domain**

DNS Typical Process

1. Client checks its own DNS cache
2. **Recursive DNS nameserver** checks its own DNS cache
3. Recursive DNS nameserver queries **authoritative DNS nameservers**



DNS Attacks and Security

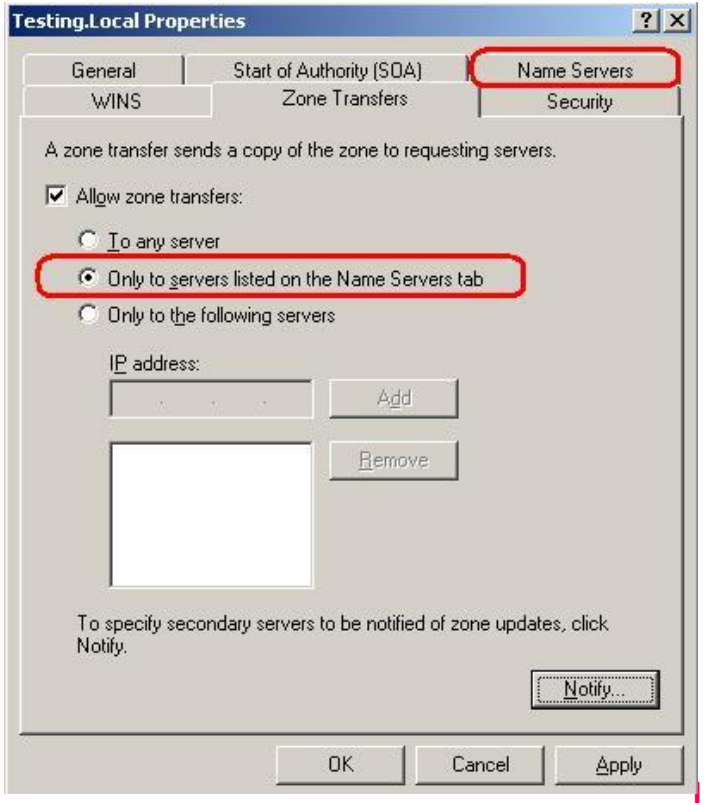
checklist time!

DNS Zone Transfers

`nslookup` can be used to obtain DNS records from DNS servers via zone transfers.

Zone transfers can expose a lot of information about your network.

You should limit who can ask for a zone transfer, or iust disable them completely.

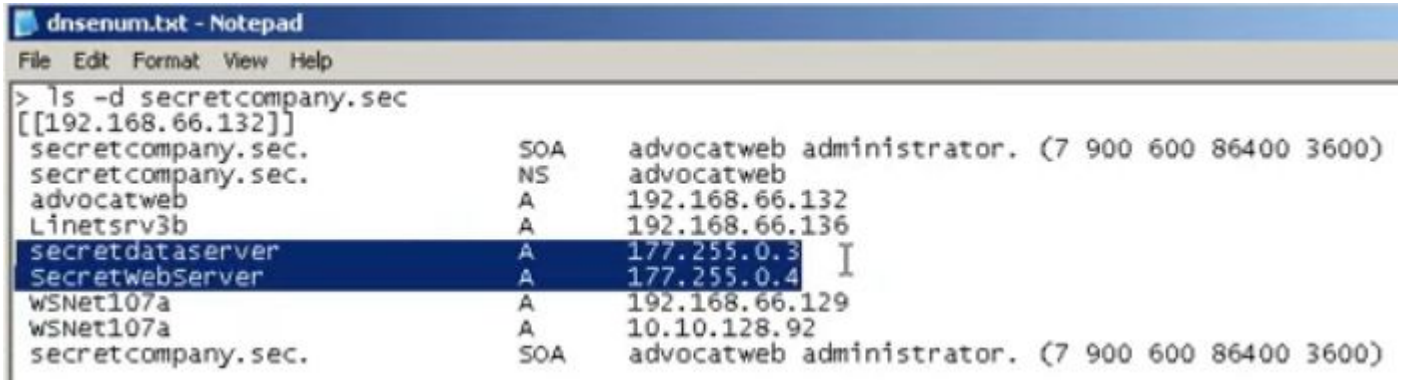


DNS Zone Transfer Attack

Recon in Play

DNS zone transfers can obtain extra information where scanning would be impossible to detect.

Highlighted section implies critical systems that are not in the same network, but DNS records expose them



DNS Dynamic Updates
Zone Poisoning

Hackers could pollute the DNS records if they are in a network where DNS servers allow nonsecure dynamic updates

Updates can be sent by a spoofed IP address to point a domain name to another IP address or create a fake malicious record

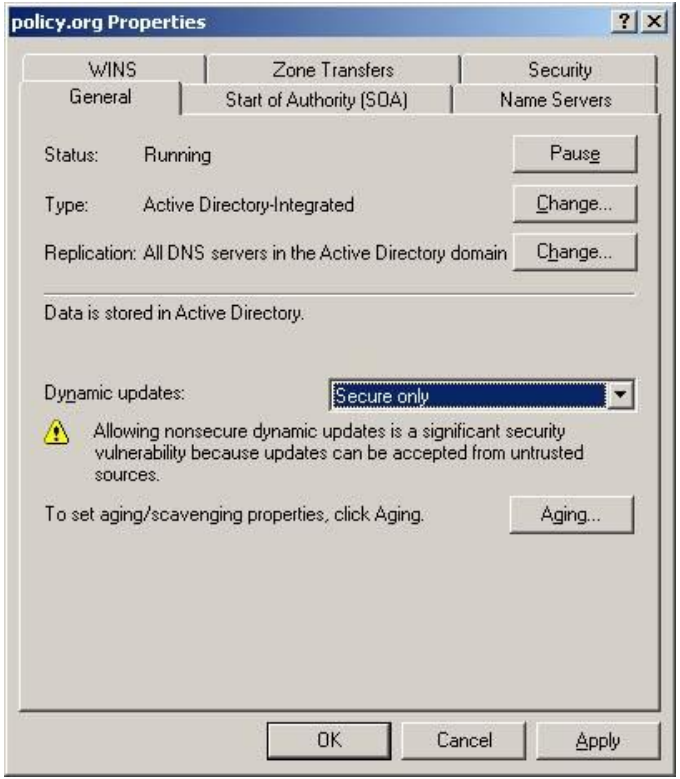
DNS Zone Poisoning
Mitigation

“Secure only” limits updates to only computers who join a domain or are in the domain (Kerberos authenticates the requests)

“None” requires manual updating of records

If only static IP addresses are used, set to this

“Nonsecure and secure” is used only if the DNS zone is not AD-integrated zone and if you want automatic updates



DNS Cache Poisoning

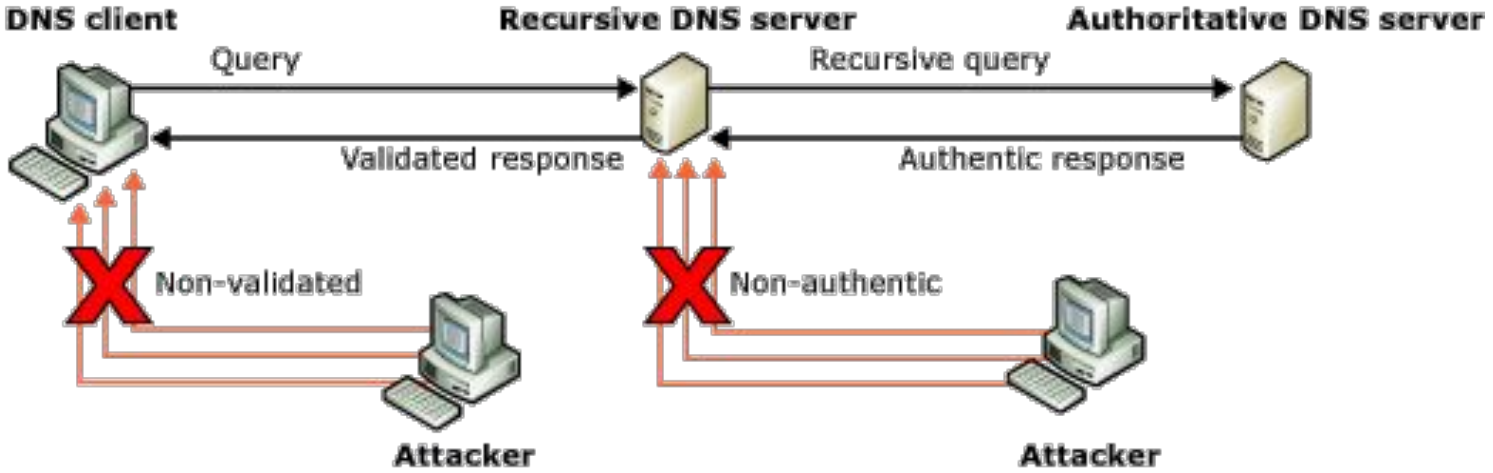
Malware can poison your DNS cache and redirect common sites to a malicious IP address instead.

View DNS resolver cache with `ipconfig /displaydns`

Run `ipconfig /flushdns` to clear your DNS cache.

Other Considerations

DNSSEC - verification (prevents spoofing)



Uses public key cryptography

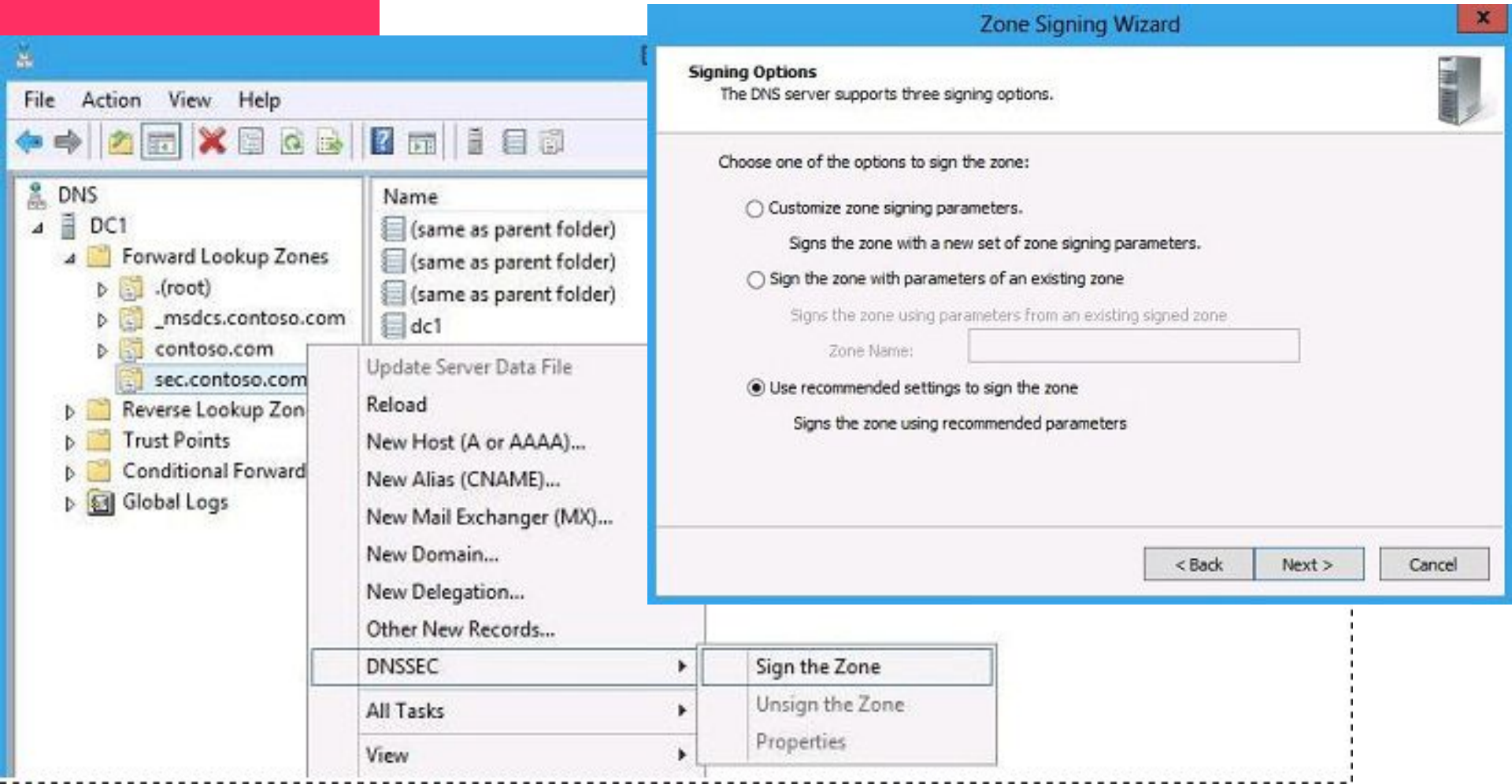
Creates DNSKEY, RRSIG and NSEC3 records

DNS DoS
Denial of Service

Allowing DNS recursion to resolve IPs outside the internal network can cause a DoS

Depends on company policy

- If need to access Internet, leave recursion enabled
- If DNS server is purely authoritative, disable recursion


File Action View Help
DNS
DC1
Forward Lookup Zones
.(root)
_msdcs.contoso.com
contoso.com
sec.contoso.com
Reverse Lookup Zon
Trust Points
Conditional Forward
Global Logs
Name
(same as parent folder)
(same as parent folder)
(same as parent folder)
dc1
Update Server Data File
Reload
New Host (A or AAAA)...
New Alias (CNAME)...
New Mail Exchanger (MX)...
New Domain...
New Delegation...
Other New Records...
DNSSEC
All Tasks
View
Sign the Zone
Unsign the Zone
Properties
Zone Signing Wizard
Signing Options
The DNS server supports three signing options.
Choose one of the options to sign the zone:
Customize zone signing parameters.
Signs the zone with a new set of zone signing parameters.
Sign the zone with parameters of an existing zone
Signs the zone using parameters from an existing signed zone
Zone Name:
Use recommended settings to sign the zone
Signs the zone using recommended parameters
< Back
Next >
Cancel

DNS Lab

Practice what you have learned



tOrange.biz

DNS Lab!

Via terminal:

1. Access google.com in a web browser via its IP address
2. Find the FQDN of 1.0.0.1
3. Obtain the MX records of google.com
4. Zone transfer the entire google.com zone to a file

On image:

5. Limit zone transfers
6. Allow only secure dynamic updates
7. Flush DNS cache

*use google to figure out the commands you need!

DNS Lab Guide
Exercise 1

Access google.com in a web browser via its IP address

1. Ping Google's domain name to get their IP address

```
ping google.com
```

   a. Alternatively, you can use nslookup

```
nslookup google.com
```

2. Enter the IP address in a web browser

IP result may vary, but should be within this:

172.217.0.0\19

Use a subnet calculator to figure out what this means!

DNS Lab Guide
Exercise 2

Find the FQDN of 1.0.0.1

1. Use nslookup

```
nslookup 1.0.0.1
```

one.one.one.one

DNS Lab Guide
Exercise 3

Obtain the MX records of google.com

1. Use nslookup

```
nslookup -type=mx google.com
```

OR

```
nslookup
> set type=mx
> google.com
```

```
Non-authoritative answer:
google.com      MX preference = 40, mail exchanger = alt3.aspmx.l.google.com
google.com      MX preference = 20, mail exchanger = alt1.aspmx.l.google.com
google.com      MX preference = 50, mail exchanger = alt4.aspmx.l.google.com
google.com      MX preference = 10, mail exchanger = aspmx.l.google.com
google.com      MX preference = 30, mail exchanger = alt2.aspmx.l.google.com
```

DNS Lab Guide
Exercise 4

Zone transfer the entire google.com zone to a file

1. Get the nameservers of Google

nslookup

> set type=ns

> google.com

2. Set nameserver

> server <nameserver>

3. Set query to get all information

> set type=any

4. Zone transfer to a file

> ls -d google.com > output.txt

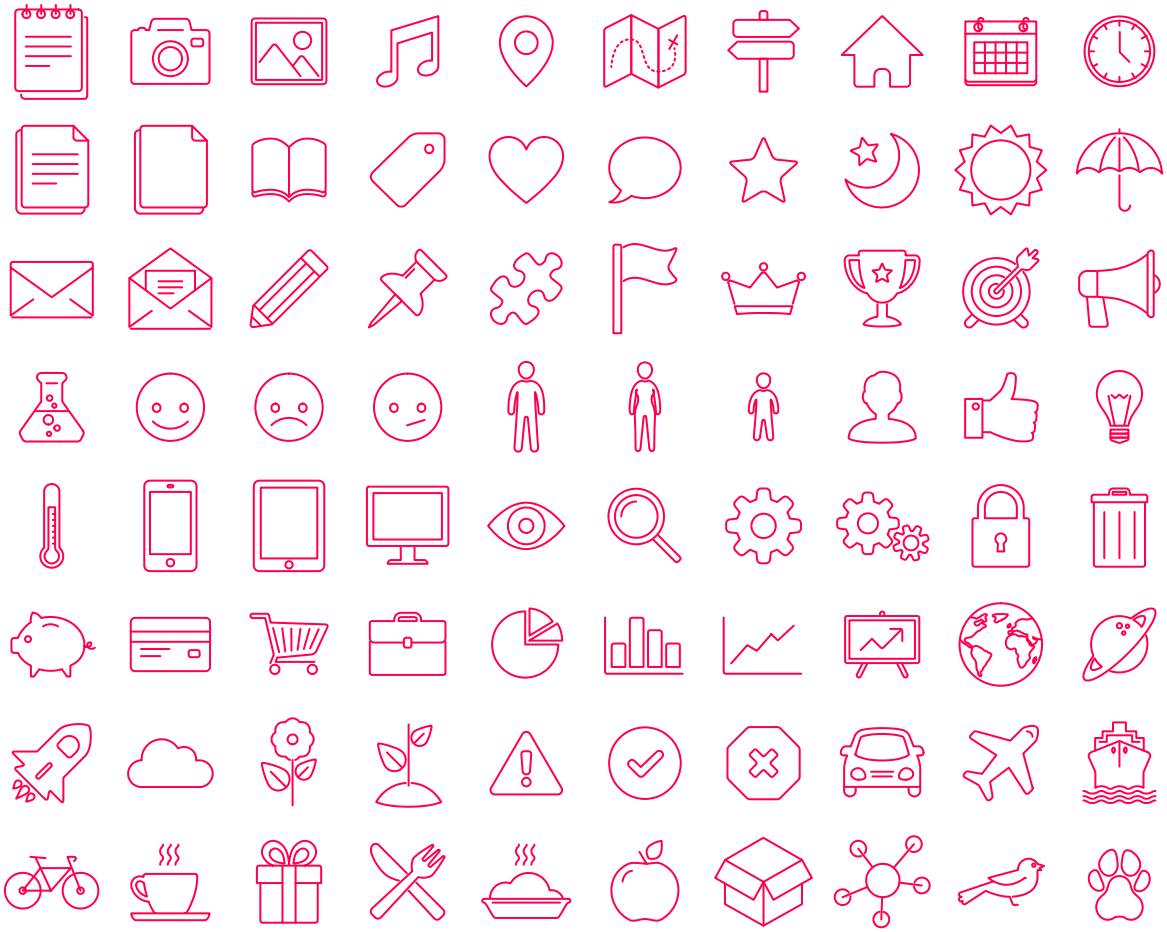
You won't be able to zone transfer because the nameserver has probably denied zone transfers from anyone except a select few trusted DNS servers.

If DNS ain’t happy, ain’t nobody happy.

Credits

Special thanks to all the people who made and released these awesome resources for free:

- Presentation template by SlidesCarnival



SlidesCarnival icons are editable shapes.

This means that you can:

- Resize them without losing quality.
- Change line color, width and style.

Isn't that nice? :)

Examples:



